Course outline

1. Linear ordinary differential equations (ODEs) and initial value problems (IVPs)
2. Power series solutions to ODEs
3. Laplace transforms
4. Fourier series
5. Boundary value problems (BVPs) and linear partial differential equations (PDEs)
6. Fourier transforms

Topic 6: Fourier transforms

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How Topic 6 builds on previous topics:

- We will derive the Fourier transform from Fourier series (cf. Topic 4)
- We will find similarities to the Laplace transform (cf. Topic 3)
- We will apply Fourier transforms to ODEs and PDEs (cf. Topics 1+5)

Topic 6: Fourier transforms

Leaning outcomes:

1. How to evaluate Fourier transforms
2. Understand basic properties of Fourier transforms and how to apply them
3. Know when and how to apply Fourier transform to ODEs and PDEs

Lecture 28

In Lecture 28, we will:

1. Define Fourier transform and contrast with Laplace transform
2. Calculate FT of example functions
3. Introduce inverse FT
4. Connect FT with Fourier series
5. Outline theorem for existence of FT and inverse FT

Recap (Topic 3): The Laplace transform

Definition (Laplace transform)

The *Laplace transform*, $\mathscr{L}$, of the function $f(t)$ is

$$\mathscr{L}\{f\} = \int_0^\infty f(t)\,\mathrm{e}^{-s\,t}\,\mathrm{d}t = F(s)$$

- Maps from t-space to s-space, where s is a complex-valued variable
- It is a linear transformation

Basic properties

Since it is a *linear transformation*:

1. $\mathscr{L}\{f+g\} = \mathscr{L}\{f\} + \mathscr{L}\{g\} = F(s) + G(s)$
2. $\mathscr{L}\{\beta f\} = \beta\mathscr{L}\{f\} = \beta F(s)$ where β is a constant

The Fourier transform

Definition (Fourier transform)

The *Fourier transform*, $\mathscr{F}$, of the function $f(x)$ is

$$\mathscr{F}\{f\} = \int_{-\infty}^{\infty} f(x)\,\mathrm{e}^{-\mathrm{i}\,k\,x}\,\mathrm{d}x = \widehat{f}(k)$$

- Maps from x-space to k-space
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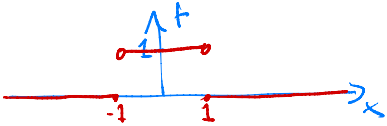
Exercise 6.1: Calculate the Fourier transform of

$$f(x) = \begin{cases} 1 & |x| < 1 \\ 0 & \text{otherwise} \end{cases}$$



Exercise 6.1: Calculate the Fourier transform of

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$$\widehat{f}(k) = \int_{-\infty}^{\infty} f(x)\, \mathrm{e}^{-\mathrm{i}\,k\,x}\, \mathrm{d}x$$

$$= \int_{-1}^{1} e^{-ikx}\, dx = \frac{-1}{ik}\left[e^{-ikx}\right]_{x=-1}^{1} = \frac{+1}{ik}\left(e^{ik} - e^{-ik}\right)$$

$$= \frac{2}{k}\,\frac{1}{2i}\left(e^{ik} - e^{-ik}\right)$$

$$= \frac{2}{k}\sin(k)$$

$$= 2\,\underbrace{\mathrm{sinc}(k)} = \begin{cases} \dfrac{\sin(k)}{k} & k \neq 0 \\ 1 & k = 0 \end{cases}$$

Exercise 6.2: Calculate the Fourier transform of $f(x) = \mathrm{e}^{-|x|}$

$$\hat{f}(k) = \int_{-\infty}^{\infty} f(x)\,\mathrm{e}^{-\mathrm{i}kx}\,\mathrm{d}x = \int_{-\infty}^{\infty} e^{-|x|} e^{-ikx}\,dx = \int_{-\infty}^{0} e^{x} e^{-ikx}\,dx + \int_{0}^{\infty} e^{-x} e^{-ikx}\,dx$$

$$= \lim_{a\to\infty} \int_{-a}^{0} e^{x-ikx}\,dx + \lim_{b\to\infty} \int_{0}^{b} e^{-x-ikx}\,dx$$

$$= \lim_{a\to\infty} \frac{1}{1-ik}\Big[e^{(1-ik)x}\Big]_{-a}^{0} + \lim_{b\to\infty} \frac{1}{-(1+ik)}\Big[e^{-(1+ik)x}\Big]_{0}^{b}$$

$$= \frac{1}{1-ik} + \frac{1}{1+ik} = \frac{1+\cancel{ik}+1-\cancel{ik}}{(1+ik)(1-ik)} = \frac{2}{1+k^2}$$

Recap (Topic 3): Inverse Laplace transform

Theorem

- Let f be a continuous function on $t \in [0, \infty)$ that has a Laplace transform, F
- Then the *inverse Laplace transform* also exists, i.e.,

$$\mathscr{L}^{-1}\{F\} = f \quad \text{for all } t \in [0, \infty)$$

The inverse Laplace transform is linear:

1. $\mathscr{L}^{-1}\{F + G\} = \mathscr{L}^{-1}\{F\} + \mathscr{L}^{-1}\{G\} = f + g$
2. $\mathscr{L}^{-1}\{\beta F\} = \beta \mathscr{L}^{-1}\{F\} = \beta\, f$ where β is a constant

*A formula for the inverse exists but it requires advanced complex analysis

Inverse Fourier transform

If the FT exists, then we introduce an inverse Fourier transform, $\mathscr{F}^{-1}$, such that

$$\mathscr{F}^{-1}\Big(\mathscr{F}(f)\Big) = \mathscr{F}^{-1}\Big(\widehat{f}\Big) = f(x)$$

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Exercise 6.3: Find the inverse ~~LT~~ FT of $\mathrm{e}^{-\mathrm{i}\,k\,a}$

Note that

$$\mathscr{F}\{\delta(x-a)\} = \int_{-\infty}^{\infty} \delta(x-a)\,\mathrm{e}^{-\mathrm{i}\,k\,x}\,\mathrm{d}x = \mathrm{e}^{-\mathrm{i}\,k\,a}$$

$$\therefore \quad F^{-1}\left(e^{-ika}\right) = F^{-1}\left(F\left(\delta(x-a)\right)\right) = \delta(x-a)$$

Inverse Fourier transform

Definition (Inverse Fourier transform)

The *inverse Fourier transform*, $\mathscr{F}^{-1}$, of the function $\widehat{f}(k)$ is

$$\mathscr{F}^{-1}\{\widehat{f}\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} \widehat{f}(k)\,\mathrm{e}^{+\mathrm{i}\,k\,x}\;\mathrm{d}k$$

- Maps from k-space to x-space
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Existence

Theorem

Let $f : \mathbb{R} \to \mathbb{C}$ be a piecewise continuous function with finitely many critical points and finitely many discontinuities on every closed interval within its domain such that

$$\int_{-\infty}^{\infty} |f(x)| \, \mathrm{d}x \quad \text{converges} \quad (\textit{absolutely integrable})$$

Then, the Fourier transform of $f(x)$ exists and

1. if $f(x)$ is continuous at x, then $\mathscr{F}^{-1}\Big(\mathscr{F}(f)\Big) = f(x)$
2. if $f(x)$ is discontinuous at x, then

$$\mathscr{F}^{-1}\Big(\mathscr{F}(f)\Big) = \frac{1}{2}\left(\lim_{\xi \to x^-} f(\xi) + \lim_{\xi \to x^+} f(\xi)\right)$$

Fourier series $\to$ Fourier Transforms

Definition (complex Fourier series, from Topic 4)

- The function $f(x)$ on $(-L, L]$ has the *complex Fourier series*

$$f(x) = \sum_{n=-\infty}^{n=\infty} c_n \mathrm{e}^{\mathrm{i}k_n x} \quad \text{where} \quad c_n = \frac{1}{2L}\int_{-L}^{L} f(x)\mathrm{e}^{-\mathrm{i}k_n x}\mathrm{d}x \quad \text{and} \quad k_n = \frac{n\pi}{L}$$

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{ik_n x} = \frac{1}{2\pi}\sum_{n=-\infty}^{\infty} 2\pi c_n e^{ik_n x}$$

$$= \frac{1}{2\pi}\sum_{n=-\infty}^{\infty} \underbrace{\pi/L}_{k_{n+1}-k_n = \Delta k} \, 2L c_n e^{ik_n x}$$

Now $\quad 2Lc_n = \cancel{\frac{2L}{2L}} \int_{-L}^{L} f(x) e^{-ik_n x}\, dx = \hat{f}(k_n)$

$$\Rightarrow \quad f(x) = \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} \hat{f}(k_n) e^{ik_n x} \Delta k$$

Take limit $L \to \infty$ then

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(k) e^{ikx} dk$$

and $$\hat{f}(k) = \int_{-\infty}^{\infty} f(x) e^{-ikx} dx$$

WARNING: Different definitions

There are different definitions of the Fourier transform in use.

$$\widehat{f}(k) = \int_{-\infty}^{\infty} f(x)\mathrm{e}^{-ikx}\mathrm{d}x \qquad f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \widehat{f}(k)\mathrm{e}^{ikx}\mathrm{d}k$$

$$\widehat{f}(k) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(x)\mathrm{e}^{-ikx}\mathrm{d}x \qquad f(x) = \int_{-\infty}^{\infty} \widehat{f}(k)\mathrm{e}^{ikx}\mathrm{d}k$$

$$\widehat{f}(k) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(x)\mathrm{e}^{-ikx}\mathrm{d}x \qquad f(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \widehat{f}(k)\mathrm{e}^{ikx}\mathrm{d}x$$

$$\widehat{f}(k) = \int_{-\infty}^{\infty} f(x)\mathrm{e}^{-2\pi ikx}\mathrm{d}x \qquad f(x) = \int_{-\infty}^{\infty} \widehat{f}(k)\mathrm{e}^{2\pi ikx}\mathrm{d}k$$

Also, often k (wavenumber) is replaced with p (momentum) and when we have functions of time we usually use ω (angular frequency) or f (frequency).